DiaLLM: An Investigation into the Robustness–Generation Gap in English Dialect Adaptation

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Abstract

Large language models increasingly *understand* dialectal English, yet still *produce* only standard, US-leaning English, leaving dialectal generation, the harder half of the problem, largely unaddressed. We introduce **DiaLLM**, which continually pretrains three open-weight language model families on the International Corpus of English and applies implicit and explicit post-training paradigms, each combined with three model alignment strategies, giving the first controlled comparison of these components across Australian, Indian, and Northern British English. Our results reveal a *robustness–generation gap*: benchmarks are shaped by continual pretraining and SFT, while alignment visibly reshapes generation in ways benchmarks do not capture. Explicit variety-targeted adaptation produces output reliably recognised as dialectal and judged more dialectal than broad alignment, yet where human judgement was directly assessed, the method that most aggressively optimises the dialectal reward is not the one judged most dialectal. Independent linguistic analysis corroborates this reward–quality gap, most clearly on two of the three families. No single alignment method dominates, and closing the gap will require richer reward designs and continued investment in dialectal resources. We release all code, checkpoints, and preference datasets.

1 Introduction

Large language models (LLMs) achieve strong performance across a range of NLP tasks (Wang et al., 2024; Touvron et al., 2023), but often falter on dialectal and non-standard varieties that diverge from standardised norms (Joshi et al., 2025; Blodgett et al., 2016; Hofmann et al., 2024; Mire et al., 2025). These disparities reflect structural imbalances in training data: large-scale corpora privilege majority language forms, marginalising non-standard varieties (Bender et al., 2021; Joshi et al., 2020). While domain adaptation has received substantial attention (Gururangan et al., 2020), robustness to dialectal variation within a single language remains comparatively underexplored. Existing work is concentrated heavily on AAVE and code-switching (Blodgett et al., 2016; Sap et al., 2019; Hofmann et al., 2024), while regional varieties such as Australian, Indian, and Northern British English receive substantially less attention at the alignment stage. Methods that do address dialectal variation, such as TADA (Held et al., 2023) and HyperLoRA (Xiao et al., 2023), do so via dialect adapters or low-rank components targeting NLU robustness under dialectal *input*. LoRDD (Srirag et al., 2025a) similarly employs a low-rank dialect adapter for decoder models in a task-specific setting. None of them investigates *full-pipeline adaptation* across pretraining, fine-tuning, and alignment, nor whether models can produce dialectally appropriate *output*.

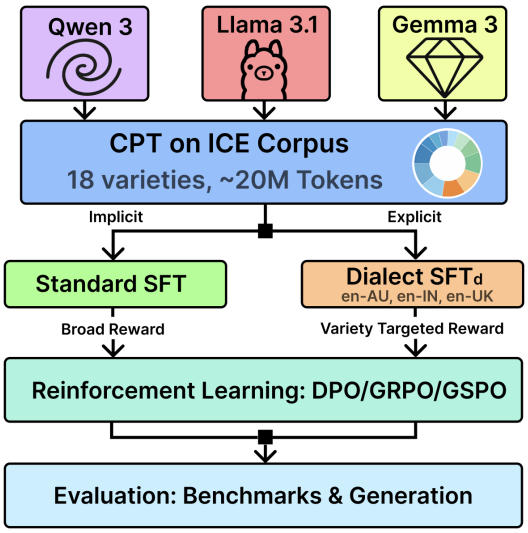


Figure 1: Overview of the **DiaLLM** pipeline: continual pretraining on ICE followed by either *implicit adaptation* (standard SFT + alignment) or *explicit adaptation* (dialectal SFT + variety-targeted alignment), with DPO, GRPO, and GSPO compared across both paradigms.

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Dialectal generation (producing a language variety on output, not merely tolerating it on input) is a critical part of the problem that current methods do not address. Consider three users chasing a late parcel:

> *Indian English:* My parcel is not **yet coming**; kindly **do the needful** and **prepone** the delivery.
>
> *Northern British English:* **Me** parcel's not arrived; **summat** were up **wi'** the courier.
>
> *Australian English:* Me parcel still hasn't turned up, **reckon** it's good this **arvo, eh?**

A modern LLM will *understand* all three, yet answer each in flattened Standard American English (Fleisig et al., 2024). A model that understands every variety but speaks only one has not solved the dialect problem; it has hidden half of it. This gap matters not because every speaker necessarily requires a response in their own variety, but because they should have the option.

We introduce **DiaLLM**, a dialect adaptation framework that continually pretrains three open-weight LLM families (Llama-3.1-8B, Qwen3-8B, Gemma-3-4B-IT) on the International Corpus of English (ICE; Greenbaum and Nelson 1996), spanning 18 English varieties. As shown in Figure 1, from each CPT checkpoint we pursue two post-training paradigms: *implicit adaptation* applies standard instruction-tuning and broad alignment; *explicit adaptation* uses dialectally-perturbed supervision and variety-targeted alignment. Across both paradigms, we compare Direct Preference Optimisation (DPO), Group Relative Policy Optimisation (GRPO), and Group Sequence Policy Optimisation (GSPO). We treat the relative effectiveness of each pipeline component and alignment method as an empirical question rather than assuming any single recipe is best.

Our results reveal a *robustness–generation* gap: benchmark performance is driven primarily by CPT and SFT, while alignment visibly shapes generation in ways benchmarks do not capture. Explicit variety-targeted adaptation is reliably recognised as dialectal and judged more dialectal than broad alignment by both human annotators and LLM judges. Yet, the method that most aggressively optimises the dialectal reward is not the one judged most dialectal. Our linguistic analysis confirms that this method produces the fewest recognisable dialectal markers across all three families, showing a mismatch between feature-density reward and perceived dialectal quality.

We make three primary contributions:

1. **DiaLLM**, a full-pipeline dialect adaptation framework with a controlled comparison of continual pretraining, supervised fine-tuning, and three alignment methods (DPO, GRPO, GSPO) across two post-training paradigms and three open-weight LLM families. We release all code (including a linguistic-analysis toolkit), model checkpoints, and dialectal preference datasets[1].

2. We establish a **robustness–generation gap**: benchmark performance is set by CPT and SFT, whereas alignment moves benchmarks only marginally, yet visibly reshapes generation, showing benchmarks alone misjudge what alignment does.

3. We expose a **reward–quality gap**: although explicit, variety-targeted adaptation is reliably recognised as dialectal and judged more dialectal than broad alignment, the method that most aggressively optimises the dialectal reward (GRPO) is *not* the one judged most dialectal by LLM judges or human annotators on Llama-3.1-8B, where this evaluation is conducted; an independent linguistic analysis across all three families corroborates this finding.

2 Related Work

LLMs systematically underperform on dialectal and non-standard English: standard NLP tools fail on AAVE (Blodgett et al., 2016), dialectal text is disproportionately flagged as toxic (Sap et al., 2019), and LLMs encode racio-linguistic stereotypes and generate lower-quality outputs for dialectal prompts even after alignment (Hofmann et al., 2024; Mire et al., 2025), with reward models themselves steering outputs toward White Mainstream English (Mire et al., 2025). These gaps are documented at scale across 281 varieties using VALUE (Ziems et al., 2022), BESSTIE (Srirag et al., 2025b), and DialectBench (Faisal et al., 2024). Maheshwari et al. (2026) similarly target robustness via LoRA and mixture-of-experts

[1] https://github.com/jordanpainter/diallm

adapters, but for classification rather than generation.

Methods that do address dialectal variation—TADA (Held et al., 2023) and HyperLoRA (Xiao et al., 2023)—do so via dialect adapters or low-rank components targeting NLU robustness under dialectal *input*; LoRDD (Srirag et al., 2025a) similarly employs a low-rank dialect adapter for decoder models in a task-specific setting. None of these existing works investigates full-pipeline adaptation across pretraining, fine-tuning, and alignment paradigms, nor whether models can produce dialectally appropriate *output*.

Continual pretraining on in-domain text is a well-established adaptation foundation (Gururangan et al., 2020), applied across medical, legal, and multilingual settings (Xie et al., 2025; Huang et al., 2023); Khan et al. (2025) show dialect gaps in Québec French can be narrowed via CPT with limited benchmark degradation. For post-training, DPO (Rafailov et al., 2023), GRPO (Shao et al., 2024), and GSPO (Zheng et al., 2025) offer contrasting approaches to preference optimisation; De Langis et al. (2024) demonstrate that RL with automatic linguistic reward signals can steer generation toward stylistically defined targets—a principle we extend to dialect-specific features. None of these methods have been systematically compared for dialect adaptation.

3 Methodology

3.1 Base Models

We apply our dialect adaptation pipeline to three open-weight LLM families spanning the 4B–8B parameter range: **Llama-3.1-8B** (Grattafiori et al., 2024), **Qwen3-8B** (Yang et al., 2025), and **Gemma-3-4B-IT** (Team et al., 2025)[2]. Selecting architecturally diverse models allows us to assess whether gains from dialect adaptation are robust across training regimes and vocabulary choices, rather than specific to a single architecture.

3.2 Continual Pretraining

Corpus: We continually pretrain all models using the International Corpus of English (ICE), a collection of comparable corpora from over 25 English-speaking regions. Each variety contains one million words of spoken and written language sampled using consistent sociolinguistic criteria, making ICE uniquely suited for this task: it is the only available multi-variety corpus spanning both inner and outer circle varieties at a usable scale, with no comparable resource covering this range of regional Englishes.

For this work, we include 18 ICE varieties with cleaned, digitised data available.[3] Access to further varieties was sought but could not be obtained, either because the relevant ICE teams did not respond to requests or because the data was not in a usable digitised form. The corpus covers a broad range of regional, cultural, and postcolonial Englishes, including Hong Kong, Nigeria, India, Jamaica, and others. In total, our training corpus comprises approximately 20 million tokens.

Pretraining setup: We continually pretrain each selected model on the cleaned ICE corpus before any supervised or preference-based post-training using GaLore (Zhao et al., 2024), enabling memory-efficient full-parameter optimisation. CPT is implemented as causal language modelling over the cleaned ICE text; full training details are provided in Appendix C.

3.3 Post-Training Dataset

For post-training, we use the preference dataset introduced by Cui et al. (2024), which contains high-quality prompt-response pairs annotated with binary preferences over completions. For the dialect-specific training thread, we convert the preferred completions into dialectal variants using the Multi-VALUE library (Ziems et al., 2023), targeting three English varieties: Australian English, Indian English, and Northern British English. Multi-VALUE applies transformations based on eWAVE features (Kortmann et al., 2020), which constitute the only feature-level inventory of morphosyntactic variation spanning all English varieties considered in this work. This makes eWAVE a natural foundation for both dialect perturbation and reward specification. During conversion, code blocks are automatically detected and preserved verbatim. The resulting datasets contain approximately 11.8k, 15.4k, and 18.4k preference pairs for Australian, Northern British, and Indian English, respectively. Differences in dataset size arise from unsuccessful conversions produced by Multi-VALUE.

[2]For Gemma-3-4B-IT, we initialise CPT from the instruction-tuned variant rather than the base model, following poor performance in preliminary experiments with the base checkpoint; the remaining pipeline is identical.

[3]See Appendix A for preprocessing and filtering details. We could only obtain permissions to use the corpus for this set.

3.4 Alignment Strategy

From each CPT checkpoint, we pursue two dialect adaptation paradigms: the **implicit thread** (broad training, with no variety targeting) and the **explicit thread** (single-variety targeting throughout), as illustrated in Figure 1. We treat each stage as serving a distinct role in the pipeline: continual pretraining introduces dialectal exposure that is otherwise absent from models trained predominantly on standard English corpora; post-training methods can shape output distributions but cannot introduce dialectal signal that is not already latent in the model's weights; supervised fine-tuning restores general instruction-following behaviour.

3.4.1 Supervised Fine-Tuning

Both adaptation threads apply SFT from the CPT checkpoint, training only on chosen completions in a single-turn instruction format with early stopping based on held-out evaluation loss. Full hyperparameters are reported in Appendix C.

The **implicit thread** fine-tunes on the standard *ultrafeedback-binarized-preferences* corpus, providing a general instruction-following objective with no explicit dialectal signal. The **explicit thread** fine-tunes instead on the dialect-specific dataset for the target variety, so the chosen completions already contain Multi-VALUE dialectal features. No general SFT stage intervenes between CPT and dialect SFT in the explicit thread, preserving the dialectal signal introduced by CPT.

3.4.2 Direct Preference Optimisation

DPO (Rafailov et al., 2023) is an offline method that optimises directly from contrastive pairs without querying a reward model. Preference pairs are drawn from the dialectal datasets in Section 3.3: chosen completions are Multi-VALUE dialectal variants; rejected completions are the original standard English forms. In the implicit thread, DPO trains on the combined three-variety dataset; the three varieties are pooled rather than targeted individually, preserving the variety-agnostic objective of this thread. In the explicit thread, only the target-variety subset is used.

3.4.3 Dialect Feature Classifier

To provide a reward signal for online alignment, we train a multi-label classifier to detect 135 morphosyntactic and lexical features of dialectal English derived from the eWAVE typological database (Kortmann and Lunkenheimer, 2013). eWAVE provides the only structured feature-level typology of morphosyntactic variation spanning our three target varieties, making it the sole available basis for a linguistically interpretable reward signal at this level of dialectal specificity. Full architecture, training, and calibration details are given in Appendix D. The classifier's logits are passed through a sigmoid and the resulting probabilities are aggregated as $\phi_{\text{dial}}(y) = \log(1 + \sum_{i \in \mathcal{F}} \sigma(\text{logit}_i))$, where $\mathcal{F}$ is the active feature set: all 135 features in the implicit thread, or the attested variety-specific subset in the explicit thread (en-AU: 20 features; en-UK: 30 features; en-IN: 38 features).

3.4.4 Reward Formulation

Both GRPO and GSPO optimise a shared reward combining dialectal feature density and meaning preservation:

$$R = \lambda \cdot \phi_{\text{dial}}(y) + \tfrac{1-\lambda}{2} \cdot \phi_{\text{comet}}(y, \hat{y}) + \tfrac{1-\lambda}{2} \cdot \phi_{\cos}(y, \hat{y}) \quad (1)$$

$\phi_{\text{dial}}(y)$ is the log-sum dialectal reward from the feature classifier (Section 3.4.3). $\phi_{\text{comet}}(y, \hat{y})$ and $\phi_{\cos}(y, \hat{y})$ measure semantic fidelity against the SFT chosen completion $\hat{y}$ using COMET (Rei et al., 2022) and cosine similarity over `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019) respectively. We fixed $\lambda = 0.80$ across all model families and alignment methods; full selection details are given in Appendix C. All components are z-score normalised.

3.4.5 GRPO and GSPO

GRPO (Shao et al., 2024) generates a group of rollouts per prompt, estimates a baseline from the group mean reward, and computes token-level policy gradient updates without a value network. GSPO (Zheng et al., 2025) replaces token-level importance sampling with sequence-level reweighting, assigning credit by the full-sequence probability ratio under the current and reference policies, avoiding the bias introduced by per-token clipping. Both use four completions per prompt and the reward in Section 3.4.4.

4 Experiment Setup

4.1 General Benchmarks

We evaluate general capability using BBH (Suzgun et al., 2023), GPQA (Rein et al., 2024), and GLUE (Wang et al., 2018), implemented via the *EleutherAI/lm-evaluation-harness* (Gao et al., 2023) with unmodified task templates.

	Qwen3-8B				Llama-3.1-8B				Gemma-3-4B-IT			
Benchmark	SFT_d	DPO	GRPO	GSPO	SFT_d	DPO	GRPO	GSPO	SFT_d	DPO	GRPO	GSPO
en-AU (Australian English)												
GLUE	61.59	61.08	61.60	**61.65**	46.26	47.11	47.08	**47.31**	52.69	**53.39**	51.98	52.95
VALUE	**55.94**	50.00	55.04	55.59	45.42	**47.06**	45.75	45.61	50.59	47.41	49.28	**50.60**
BST-Sent	**90.02**	89.57	89.93	89.92	87.31	87.26	**87.35**	87.21	**84.91**	84.73	**84.91**	84.76
BST-Sarc	37.07	**37.36**	36.84	36.52	39.45	**41.18**	40.54	38.95	27.31	**31.86**	28.90	28.30
DID	**67.11**	65.44	66.61	64.94	67.95	64.44	67.11	**69.28**	35.56	**35.73**	35.56	35.56
en-IN (Indian English)												
GLUE	62.71	**63.03**	62.50	62.84	54.27	**56.45**	53.94	53.88	57.15	57.63	57.12	**58.38**
VALUE	52.92	50.41	**52.94**	52.90	48.32	**49.34**	47.88	48.01	50.21	49.49	50.68	**50.78**
BST-Sent	89.75	89.61	89.78	**90.01**	81.52	80.95	**82.64**	82.09	77.50	75.53	78.97	**79.11**
BST-Sarc	39.27	38.50	**39.43**	38.87	36.36	**37.68**	35.47	35.27	32.79	32.54	32.88	**33.22**
DID	**71.45**	70.12	70.79	70.45	64.78	**65.78**	64.11	63.44	13.02	13.02	13.02	**13.36**
en-UK (Northern British English)												
GLUE	63.48	63.42	**63.99**	63.72	**54.05**	53.77	53.95	53.78	56.03	55.35	55.45	**56.42**
VALUE	58.27	54.59	**58.50**	58.49	45.67	**48.35**	45.56	45.59	**48.93**	46.36	48.37	48.61
BST-Sent	87.92	**88.01**	**88.01**	87.99	79.45	**80.83**	80.11	79.89	82.53	79.31	82.23	**83.49**
BST-Sarc	35.44	34.97	35.28	**35.48**	34.96	34.58	**35.16**	34.73	**33.95**	31.68	33.11	32.61
DID	63.94	63.44	64.27	**64.61**	50.58	**58.60**	49.58	51.59	35.23	**35.39**	35.23	35.23

Table 1: Explicit thread alignment comparison (%). SFT_d denotes dialect-specific supervised fine-tuning; DPO, GRPO, and GSPO are applied to SFT_d with variety-targeted rewards. Bold indicates the best result within each model family per row and dialect group. BST-Sent and BST-Sarc refer to BESSTIE sentiment and sarcasm respectively.

4.2 Dialectal Benchmarks

We evaluate dialectal robustness using three benchmark suites. The dialect identification (DID) task from DIALECTBENCH (Faisal et al., 2024) requires classifying inputs as British English (en-GB), American English (en-US), or generic English (en), with higher scores indicating greater sensitivity to dialectal variation; as DID contains no en-AU or en-IN class, scores for these conditions index general dialectal sensitivity rather than variety-specific recognition. VALUE (Ziems et al., 2022) replaces GLUE inputs with AAVE variants, directly measuring performance disparities relative to the standard benchmark. BST-Sarc and BST-Sent from BESSTIE (Srirag et al., 2025b) evaluate sarcasm detection and sentiment classification across British, Indian, and Australian English, and constitute the only available coverage specific to our three target varieties among existing English dialect benchmarks.

4.3 Generation Evaluation

Benchmarks measure performance under dialectal *input* but do not reveal whether models produce dialectally marked *output*. We, therefore, conduct a dedicated generation evaluation using Llama-3.1-8B, which shows the most consistent dialect-sensitivity gains across pipeline stages and allows more detailed qualitative examination than averaging across families would permit. We focus on the explicit thread for variety-targeted generation analysis, with the broad thread also included where alignment methods are compared directly (Section 5.3).

Prompts. We construct a set of 25 open-ended casual prompts spanning five domains: casual conversation, opinion, food and lifestyle, hypothetical, and personal reflection, designed to elicit natural, conversational responses (Appendix G.2). All evaluation methods use this same prompt set.

Automatic Metrics. *Variety classifier accuracy* (Appendix F) is the proportion of responses correctly classified as the target variety by a DeBERTa-v3-base model fine-tuned on a deduplicated, curated split of BESSTIE (Srirag et al., 2025b) using an 80/10/10 stratified train/validation/test split, achieving a macro F1 of 0.766 on the held-out test set. It serves as our primary automatic proxy for holistic dialectal style.

Human Evaluation. Two native or near-native speakers per dialect (six annotators total), selected on the basis of self-reported familiarity with the target variety, complete 100 blind pairwise or three-

	Qwen3-8B						Llama-3.1-8B						Gemma-3-4B-IT					
Benchmark	Base	CPT	SFT	DPO	GRPO	GSPO	Base	CPT	SFT	DPO	GRPO	GSPO	Base	CPT	SFT	DPO	GRPO	GSPO
BBH	70.0	55.9	53.7	**74.2**	73.0	73.0	46.5	45.4	45.6	**48.0**	30.7	34.1	**48.7**	41.0	40.3	45.4	45.4	45.5
GPQA	36.6	37.2	36.2	38.4	36.4	**39.9**	**31.4**	31.0	27.8	29.3	28.8	30.3	27.8	30.3	29.5	**33.8**	29.8	30.8
GLUE	**68.4**	59.1	64.0	63.6	63.9	64.3	**52.9**	46.6	49.3	49.5	49.2	48.9	**62.8**	50.6	50.8	51.0	52.6	51.1
VALUE	56.1	50.3	65.6	64.8	**66.2**	66.0	43.2	38.8	43.8	**44.1**	43.7	42.8	**52.8**	42.3	43.7	45.1	45.5	43.4
BST-Sent	86.1	89.9	**90.2**	90.1	**90.2**	**90.2**	87.1	**87.3**	84.7	85.4	85.3	85.1	**85.2**	82.2	82.5	78.5	84.4	82.4
BST-Sarc	37.5	**42.8**	41.5	41.3	40.5	41.3	38.3	34.9	38.4	**39.0**	38.5	38.9	**34.4**	12.5	25.8	34.2	24.5	26.0
DID	56.4	61.3	64.9	**65.8**	64.6	**65.8**	38.9	40.6	59.3	50.4	57.1	**61.3**	**59.4**	35.1	26.4	13.7	34.6	27.7

Table 2: Implicit adaptation thread results (%). Base, CPT, and SFT are sequential pipeline stages; DPO, GRPO, and GSPO are alignment methods applied on top of the SFT checkpoint without dialect-specific targeting. **Bold** indicates the best result within each model family per row. BST-Sent and BST-Sarc refer to BESSTIE sentiment and sarcasm respectively.

way judgements structured across four annotation tasks of 25 trials each, comparing successive pipeline stages and alignment methods (detailed in Section 5.3). Trials are presented in blocks; model identities are not revealed. Tasks 1, 2, and 3 use A/B/Tie judgements; Task 4 is a forced three-way choice with no tie option.

LLM-as-a-Judge Evaluation. We also evaluate these same four annotation tasks using `microsoft/phi-4` (Abdin et al., 2024), a model outside the evaluated families to reduce architectural bias. Both conditions instruct the judge to focus on dialectal features and ignore fluency and writing quality. They differ only in the system prompt: *no-features*, where Phi-4 acts as a dialect specialist without an explicit feature list; and *eWAVE-features*, where the system prompt additionally provides the full eWAVE feature inventory for the target variety. Comparing conditions isolates the contribution of explicit linguistic knowledge to LLM dialectal judgements.

5 Results

5.1 Implicit and Explicit Adaptation

In the implicit thread (Table 2), CPT disrupts general benchmark performance, SFT accounts for the majority of subsequent recovery, and alignment produces inconsistent shifts with no method reliably improving over SFT. Qwen is largely robust to CPT and gains on several dialect-sensitive metrics, while Gemma is most affected and fails to fully recover DID after SFT. Alignment effects are model- and benchmark-specific rather than systematic: Qwen BBH under DPO not only recovers from CPT and SFT degradation but exceeds the base model, suggesting that dialectal alignment can in some conditions enhance general reasoning, though the basis for this gain is not established and it is not replicated across families. Conversely, Llama BBH regresses notably under GRPO and GSPO, and Gemma DID collapses under DPO. Llama DID under GSPO is a notable exception where alignment adds meaningfully over SFT on a dialect task.

Table 1 shows that in the explicit thread, SFT_d generally sets a strong performance baseline and alignment methods produce mostly small, inconsistent shifts. General capability benchmarks remain stable across all methods, suggesting alignment does not degrade general capability in the explicit thread. DPO shows the widest variance across varieties, with a consistent tendency to hurt on VALUE. Across varieties, no single method dominates and benchmark differences alone do not cleanly distinguish the alignment approaches, motivating the subsequent generation quality analysis in Section 5.2 onwards.

5.2 Reward Optimisation vs. Perceived Dialect Quality

We examine dialectal generation through automatic metrics and human and LLM judgements, with representative qualitative examples shown in Table 3. As a training diagnostic, dialect feature density (Appendix E) shows that GRPO most aggressively optimises the reward signal, achieving the highest eWAVE feature density across all three families. Yet variety classifier accuracy (Appendix F), an independent signal covering all three model families, favours DPO for en-IN, with SFT_d achieving the overall highest en-AU classifier accuracy. This divergence indicates that reward optimisation and holistic dialectal quality are not equivalent: the

Config	Prompt	Broad (`-all`)	Explicit (`-d`)
Model: Qwen3-8B **Variety:** en-UK **Adaptation:** GRPO	What's the best piece of advice someone's ever given you?	"Be kind to others, and treat them the way you would like to be treated."…a fundamental principle of many cultures.	…This advice has helped **us** navigate through difficult situations and make decisions that align with **me** core values.
Model: Llama-3.1-8B **Variety:** en-AU **Adaptation:** DPO	What are some small things that can brighten up a rough day?	…small things, depending on what makes you feel happy and energized…Treat yourself to a small indulgence…	…what makes you feel **like** happy…Kindness to **youself**: Treat **youself** to a small indulgence…
Model: Gemma-3-4B-IT **Variety:** en-IN **Adaptation:** GSPO	What's your take on people who are always late?	…I understand that being late can be a frustrating experience for others…it's not always appropriate to judge someone solely based on their punctuality…	…**I am understanding** that being late can be frustrating, especially where **the** punctuality is highly valued…it's not appropriate to judge or **are criticizing** them without knowing the full contexts…

Table 3: Qualitative comparison of broad (`-all`) and explicit (`-d`) model responses. Broad models produce dialectally neutral outputs; explicit models exhibit surface dialectal features (**bold**). en-UK: possessive *me*, first-person singular *us*. en-AU: *like* as discourse particle, *youself*. en-IN: stative progressive (*am understanding*), article overextension (*the* punctuality), progressive overextension (*are criticizing*). Examples are selected to illustrate clear surface dialectal features and are not claimed to be representative of all outputs at each pipeline stage.

method that maximises the training objective does not necessarily produce outputs that are more recognisably dialectal by an independent measure. An independent, reward-free linguistic analysis corroborates this pattern across all three families: the method that maximises the eWAVE reward (GRPO) produces the *fewest* surface dialectal markers and the lowest feature diversity for Llama and Qwen (Appendix I, Table 11; Figure 2). This is in part a property of the reward basis itself: eWAVE is a typological atlas documenting features that are attested across varieties, not a perceptual evaluation instrument. Some dialectal features overlap across multiple varieties, and others carry stronger associations with non-target varieties in general language use, meaning reward optimisation over this feature set can increase surface feature density without producing outputs that are holistically recognisable as the target variety. The en-UK classifier does not generalise to generated outputs (Appendix F), so en-UK generation quality is assessed via human and LLM judgements only.

5.3 Human Evaluation and LLM-as-a-Judge

Table 4 reports judgement results from human annotators (two per locale) alongside two Phi-4 judge conditions, evaluated on Llama-3.1-8B outputs throughout.

The clearest finding across evaluators is that explicit adaptation produces output that human annotators recognise as dialectally distinct. Annotators judge SFT_d as more dialectal than the instruction-tuned baseline at high rates for en-IN and en-UK (84% and 87% respectively, ties excluded), and judge explicitly variety-targeted alignment as more dialectal than broad alignment at similarly high rates (71% en-IN, 85% en-UK). Both Phi-4 conditions agree unanimously on the latter comparison across all three varieties. These results indicate that the explicit pipeline produces perceptible dialectal features, and that variety targeting produces more recognisable output than broad alignment regardless of evaluator type. En-AU is the exception: annotators show no clear judgement between SFT_d and the baseline (48%, ties excluded) and weakly judge broad alignment as more dialectal than variety-targeted in T2 (34%). This divergence is consistent with Australian English features being more subtle due to its historical context (Collins, 2012) and, consequently, less perceptible to annotators, and reinforces that explicit adaptation does not uniformly improve perceived dialectal quality across varieties.

Reward optimisation does not improve over supervised fine-tuning by any evaluator. When GRPO, the method achieving the highest automatic feature density, is compared directly against SFT_d, neither Phi-4 condition nor human annotators show a consistent judgement in either direction across varieties. The high tie rate on Task 3 for en-IN (28 of 50 judgements excluded from the win rate) reflects the difficulty of distinguishing GRPO_d from SFT_d outputs for this variety, consistent with the finding that reward optimisation does not reliably produce outputs that are more recognisably dialectal. This is the clearest expression of the reward–

Task	Dial.	Human	Phi-4 (no ft.)	Phi-4 (+eWAVE)
T1	en-AU	48% (8t)	43%	**81%**
	en-IN	**84%** (7t)	**71%**	**96%**
	en-UK	**87%** (3t)	32%	**60%**
T2	en-AU	34% (9t)	**100%**	**100%**
	en-IN	**71%** (8t)	**100%**	**100%**
	en-UK	**85%** (4t)	**100%**	**100%**
T3	en-AU	46% (24t)	33%	44%
	en-IN	55% (28t)	32%	24%
	en-UK	38% (18t)	56%	40%
T4	en-AU	GS (36%)	GS (40%)	GS (48%)
	en-IN	D (50%)	GS (36%)	D (56%)
	en-UK	D (52%)	D (44%)	D (52%)

Table 4: Generation judgement results across four tasks, three dialects, and three judge types. **T1**: Instruct vs SFT_d (% SFT_d wins, tie-excluded). **T2**: $GRPO_{all}$ vs $GRPO_d$ (% $GRPO_d$ wins, tie-excluded). **T3**: SFT_d vs $GRPO_d$ (% $GRPO_d$ wins, tie-excluded). **T4**: method judged most dialectal and vote share (forced choice); D=DPO, GS=GSPO. **Human**: pooled across 2 annotators per dialect (50 judgements per task). Parenthesised t = tied judgements excluded from win rates. **Bold**: $\geq$60%.

quality gap: alignment pushes the reward signal upward, but that movement does not translate into outputs that evaluators find more dialectally convincing than the SFT baseline alone. Linking these judgements to output features, human judgement tracks dialectal marking relative to the standard baseline but not reward density; annotators judge the more dialectally marked output as more dialectal than the instruct baseline yet judge the more natural, contracted output as more dialectal than GRPO (Appendix I, Table 12).

A direct comparison of the alignment methods shows that GRPO is judged least dialectal across all three varieties and both Phi-4 conditions, despite achieving the highest feature density. Human annotators judge DPO as more dialectal for en-UK and en-IN, with Phi-4 in broad agreement; for en-AU, judgements favour GSPO under both judge conditions. Full inter-annotator agreement statistics are reported in Appendix H.

6 Discussion

Our results highlight that dialectal robustness and generation are shaped by different pipeline components, and are not reliably co-indexed by any single method. CPT and SFT together determine the performance ceiling on standard and dialectal benchmarks; alignment operates largely within that ceiling, producing small and inconsistent benchmark shifts while visibly changing generation, a pattern confirmed generation-side by an independent linguistic analysis: surface dialectal marking is established at supervised fine-tuning and only redistributed by alignment, remaining near-absent through the base, CPT, and instruct checkpoints (Appendix I, Figure 3).

The clearest positive finding is that explicit variety targeting consistently produces output that human annotators and LLM judges recognise as dialectally distinct, and that variety-targeted alignment is judged more dialectal than broad alignment regardless of evaluator type (Task 2). En-AU is the exception, with annotators showing no clear judgement at either stage.

The reward–quality gap complicates this picture. GRPO achieves the highest eWAVE feature density across all three model families; yet on Llama-3.1-8B, where generation evaluation is conducted, it is the method judged least dialectal in direct comparison across all three varieties and both Phi-4 conditions (Task 4). When compared directly against SFT_d (Task 3), neither Phi-4 condition nor human annotators show a consistent judgement for GRPO in either direction, meaning reward optimisation does not add evaluator-perceived value over the supervised baseline. DPO, which consistently falls *below* SFT_d in feature density, achieves better variety classifier accuracy for en-IN and is judged more dialectal by human annotators for en-UK and en-IN. This inversion follows from the reward basis itself (Section 5.2): an inventory built to document attestation does not track perceptual dialectal quality, so raising feature density need not make output *more recognisably dialectal*.

Indian English proves consistently more challenging across most evaluation dimensions, likely reflecting the greater structural distance between IndE and the standard English training distribution and a denser active feature set (38 features) that creates more surface area for reward optimisation without producing recognisable output.

7 Conclusion

We presented DiaLLM, a dialect adaptation pipeline that continually pretrains three open-weight LLM families on 18 English varieties from the International Corpus of English, then applies two post-training paradigms with a systematic comparison of DPO, GRPO, and GSPO. Our goal was not to establish a single winning recipe, but to provide a controlled empirical account of what each pipeline component contributes and where current alignment methods succeed and fall short.

Two findings stand out. The first is a robustness–generation gap: benchmark performance is shaped primarily by CPT and SFT, while alignment shifts generation in ways that benchmarks do not capture. The second is a reward–quality gap: on Llama-3.1-8B, the method that most aggressively optimises the dialectal reward signal is not the one judged most dialectal by LLM judges or human annotators; automatic evaluation across all three model families is consistent with this direction, though human and LLM judgement evidence is limited to Llama-3.1-8B. Together, these findings show two things benchmarks alone would not have revealed: that alignment reshapes generation independently of its effect on benchmark scores, and that the method which most successfully optimises a dialectal reward is not the one evaluators find most convincing.

The strongest positive result is that explicit variety targeting consistently produces output that human annotators and LLM judges recognise as dialectally distinct, and that variety-targeted alignment is consistently judged more dialectal than broad alignment regardless of evaluator type. This holds consistently for en-IN and en-UK; en-AU is the exception, with annotators showing no clear judgement at either stage. No single alignment method dominates across conditions; results vary by model family, variety, and evaluation dimension, and the patterns we observe should not be treated as general prescriptions. Gemma shows greater sensitivity to CPT disruption than Llama or Qwen, and en-IN proves consistently more challenging than en-AU or en-UK across all methods.

We release all model checkpoints and dialectal preference datasets to support future work. The varieties studied here remain substantially under-resourced at the alignment stage, and the design of effective training signals for authentic dialectal generation is an open problem. DiaLLM provides a useful empirical foundation for understanding the current limits of dialect adaptation and a starting point for the work these varieties still need.

Limitations

Our evaluation has several limitations. First, VALUE was developed around AAVE and is not designed to benchmark our three target varieties directly; gains are best interpreted as evidence of broad dialectal robustness rather than variety-specific improvement. BESSTIE covers our target varieties but is limited to sentiment and sarcasm, leaving many dimensions of dialectal competence unassessed. Second, eWAVE was designed as a typological database for classifying variety inventories, not as a reward signal for generative training. Some attested features are pragmatically conditioned, tending to surface only in specific conversational registers or construction types that may not be elicited by generic prompts. Others represent population-level tendencies that individual speakers vary substantially on, making binary detection inherently noisy. Operationalising these features as a learnable reward therefore introduces a mismatch: the signal favours outputs that superficially satisfy classifier-detectable patterns over those that are authentically dialectal in ways the inventory was never designed to measure. Beyond this, our reward function and generation evaluation share the same eWAVE-derived feature inventory, so feature density gains—retained in Appendix E as a training diagnostic only—reflect alignment with the training feature space rather than independently verified authentic dialect use. The BESSTIE-trained variety classifier provides results subject to register biases in the underlying corpus, as evidenced by the en-UK classifier's near-total collapse to en-AU predictions on LLM-generated outputs (Section 5.2). Additionally, Phi-4 judge reliability was not validated against an independent human gold standard beyond the agreement analysis in Appendix H.

The ICE corpus consists primarily of formal and semi-formal texts—legal proceedings, academic writing, scripted broadcasts—and our models may not generalise to informal or social media registers. Our pipeline is also limited to English dialects; extension to other languages would require new CPT corpora, feature inventories, and evaluation resources. Deployment also requires knowing which variety to serve, at a cost that scales per variety; adaptive selection between variety-specific

and standard models remains future work. Generation evaluation is conducted on Llama-3.1-8B only; whether the qualitative findings and judgement results generalise to Qwen or Gemma remains to be established. More broadly, our pipeline depends on both ICE corpus coverage and eWAVE feature attestation, restricting applicability to comparatively well-resourced English varieties even within the language.

Human evaluation used two annotators per variety (six total). Annotators were not dialect specialists and were selected on the basis of self-reported familiarity with the target variety. All judgements were conducted blind to model identity. Overall inter-annotator agreement is modest, reflecting the inherent difficulty of perceptual dialectal judgements with a small annotator pool; full statistics are reported in Appendix H. En-AU agreement is weakest (mean AC1 0.02), consistent with the weak judgement results for that variety, and en-AU human judgement results should therefore be treated as indicative rather than conclusive.

Finally, the implicit and explicit adaptation threads differ not only in whether dialectal perturbation is applied during SFT but also in the training data itself, with the implicit thread using UltraFeedback and the explicit thread using the dialect-specific preference dataset. Observed generation differences between threads therefore cannot be attributed solely to variety targeting, and future work isolating these factors through controlled ablation would sharpen the interpretation.

Ethical Considerations

This work aims to reduce linguistic inequities by improving LLM performance on under-represented English varieties. However, explicit dialect modelling carries risks: Multi-VALUE transformations may introduce stereotypical or overgeneralised patterns, and dialect-sensitive models could be misused to infer demographic or regional identity from text. DiaLLM should be interpreted as a step toward dialectal inclusivity rather than a complete mitigation of dialect bias. All model checkpoints and datasets released with this work will include documentation cards describing intended use, known limitations, and the dialectal communities represented.

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A Corpus Details

ICE Variety	Cleaned Tokens	Retention (%)
Australia	655,336	100.0
Canada	1,439,016	38.0
Great Britain	1,002,845	100.0
Ireland	1,342,878	61.7
USA	637,093	63.5
India	1,570,211	41.5
Singapore	1,412,390	49.1
Malaysia	382,839	96.7
Nigeria	1,308,130	100.0
Philippines	1,486,351	45.3
Cameroon	393,254	74.0
East Africa	1,105,083	99.8
Gibraltar	539,240	78.3
Hong Kong	1,899,811	37.1
Jamaica	1,396,602	40.6
Scotland (spoken only)	584,070	100.0
Trinidad & Tobago	1,403,912	60.6
Uganda	1,355,122	34.3
Total (18 varieties)	**19,914,183**	**52.4**

Table 5: ICE varieties used for continual pretraining, with retained token counts after preprocessing. Raw token total: 38,012,253. Retention varies due to differences in formatting, transcription quality, and completeness.

We source data from 18 varieties in the International Corpus of English (ICE), listed in Table 5, selecting those with sufficiently clean and accessible text (i.e., minimal markup corruption and complete transcript availability). Where required, access was obtained by contacting individual ICE teams, and only varieties for which usable data was available were included. The corpus spans a diverse set of regional English varieties, including both 'Inner Circle' (e.g., Great Britain, USA) and 'Outer Circle' (e.g., India, Nigeria, Singapore) contexts.

Preprocessing removes markup, incomplete transcripts, and duplicated content. Retention rates vary substantially across corpora, reflecting differences in data quality and preprocessing requirements. For *Scotland*, only spoken data is included due to inconsistencies in the written portion. The final combined corpus contains approximately 20 million tokens. This corpus forms the sole source of data for continual pretraining across all models.

B Dialect Feature Inventory

Our dialect feature classifier is based on features derived from the eWAVE typological database, spanning 135 morphosyntactic and lexical phenomena across English varieties. These features form the basis of both the dialect density metric and the masked reward used in dialect-specific adaptation.

For reward masking, we use attested feature subsets for each target variety: 20 features for AusE, 38 for IndE, and 30 for NorthE. Table 6 presents a representative subset of features used for dialect-specific reward masking across the three target varieties: Northern British English (NorthE), Indian English (IndE), and Australian English (AusE).

For dialect-specific reinforcement learning, reward computation is restricted to feature subsets corresponding to the target variety, based on attested feature–dialect mappings.

C Hyperparameters and Infrastructure

All CPT and post-training experiments were conducted on NVIDIA A100 GPUs using a SLURM-managed cluster, with the `accelerate` library for distributed execution. All runs use bfloat16 precision, gradient checkpointing, and a fixed random seed of 1234.

Continual pretraining uses GaLore (Zhao et al., 2024) with AdamW over 64,000 steps, with a GaLore rank of 1024, scale 0.25, and a subspace update interval of 500 steps. The learning rate is 1×10^{-4} with linear warmup over 10% of training, weight decay of 0, gradient clipping at 1.0, and an effective batch size of 512.

For both SFT and SFT_d, models are trained for 3 epochs on chosen completions only in a single-turn instruction format, with a learning rate of 2×10^{-5}, cosine scheduling, and 5% warmup.

All DPO runs use 5,000 steps with the sigmoid loss objective and $\beta = 0.1$. Gemma uses AdamW at 5×10^{-6}; Llama and Qwen use 8-bit AdamW

Dialect	Feature	Area	Rtg	Example
NorthE	Object pronouns as possessives (1SG)	Pronouns	A	He's me brother
	Us for singular object referent	Pronouns	A	Show us them boots
	Was/were generalisation	Agreement	B	he were thirsty
	Nowt/owt (nothing/anything)	Adverbs	B	I've got nowt
IndE	Progressive with stative verbs	Tense/aspect	A	I'm liking this
	Definite article for indefinite	Noun phrase	A	I had the toothache
	Would for future	Modality	A	I would eat tomorrow
	No subject–aux inversion (wh-)	Complementation	A	What you are doing?
AusE	Progressive with stative verbs	Tense/aspect	B	I'm liking this
	Bare adverbs (adj as adv)	Adverbs	B	drive slow
	Discourse marker (sentence-final)	Discourse	B	it was good, like
	Like as quotative/focuser	Discourse	B	she was like…

Table 6: Representative eWAVE-derived features used in the dialect feature classifier and reward function. Ratings follow the eWAVE attestation scale: A (pervasive), B (common), C (rare), D (limited structural context); X = not applicable, ? = no data.

at 2×10^{-6}. Maximum prompt and completion lengths are 2048/256 for Gemma and Llama, and 1024/192 for Qwen, with a 5% warmup ratio.

GRPO and GSPO share the same configuration, differing only in their importance sampling level: GSPO applies it at the sequence level, GRPO at the token level. Both use 4 generations per prompt over 5,000 steps, with sampling temperature 0.7 and top-$p = 0.95$. We set $\beta = 0.02$, $\epsilon = 3 \times 10^{-4}$, and $\epsilon_{\text{high}} = 4 \times 10^{-4}$. Reward weights are $(0.8, 0.1, 0.1)$ for the dialect, COMET, and cosine components respectively; all components are z-score normalised using an exponential moving average with decay 0.99 and clipping at ± 5. Optimiser settings and sequence lengths follow the DPO configuration, with weight decay 0 and gradient clipping at 1.0.

We selected λ by running full training for Llama 3.1-8B across all implicit and explicit threads with $\lambda \in \{0.50, 0.66, 0.80\}$ and inspecting the composite reward trajectories. $\lambda = 0.50$ and $\lambda = 0.66$ showed faster saturation of the meaning-preservation components relative to the dialectal reward, while $\lambda = 0.80$ produced the most stable joint trajectory across threads. We therefore fixed $\lambda = 0.80$ for all model families and alignment methods.

D Dialect Feature Classifier Details

The classifier is a BERT-base encoder (Devlin et al., 2019) with a lightweight linear projection head, trained using binary cross-entropy loss. Training data comprises approximately 148,000 sentences derived from standard NLU benchmarks (Wang et al., 2018) and transformed into dialectal variants using Multi-VALUE (Ziems et al., 2023); feature labels are obtained deterministically from the transformation process. On a held-out test set, the classifier achieves moderate overall performance, with reliable detection of high-frequency morphosyntactic markers and weaker coverage of rare or context-dependent features.

Prior to reward computation we apply per-feature temperature calibration to the sigmoid outputs, performing an independent grid search over $T \in [0.5, 3.0]$ for each of the 135 features on a held-out validation set to minimise Expected Calibration Error (ECE). The resulting temperature distribution is centred around unity ($\mu = 0.994$, $\sigma = 0.108$), with targeted corrections for specific features: sharpening underconfident predictions (e.g., $T = 0.76$) and softening overconfident ones (e.g., $T = 1.78$). This calibration ensures that sigmoid probabilities used as reward signals are well-scaled and comparable across features.

E Dialect Feature Density Across Pipeline Stages (Training Diagnostic)

Table 7 reports dialect feature density as a training diagnostic: it characterises how each pipeline stage and alignment method optimises the reward signal, but is not used as a primary evaluation metric (see Section 4.3). Results cover all three model families, averaged across the 25 evaluation prompts.

Three observations are consistent across all three families. First, every pipeline stage contributes dialectal signal: the log-sum score increases monotonically from CPT through alignment in the broad

Family	CPT	SFT	DPO	GRPO	GSPO
Broad thread (`-all`)					
Gemma	2.474	2.817	2.912	**2.959**	2.949
Llama	2.297	2.890	2.939	**2.950**	2.911
Qwen	2.834	2.932	2.950	**2.951**	2.941
Explicit thread (avg. across en-AU, en-IN, en-UK)					
Gemma	—	1.570	1.504	**1.628**	1.617
Llama	—	1.537	1.469	**1.598**	1.552
Qwen	—	1.553	1.495	**1.590**	1.561

Table 7: $\log(1 + \sum \sigma(\text{logits}))$ over active eWAVE features across pipeline stages (training diagnostic only). Broad thread uses all 135 features; explicit thread uses variety-specific subsets. **Bold** indicates highest alignment method per row. CPT is not applicable to the explicit thread as it precedes dialect-specific branching.

thread, confirming that each component adds measurable reward-side signal. Second, the explicit thread is lower in absolute terms than the broad thread at equivalent stages, reflecting the smaller active feature subsets; the relative ordering across alignment methods is preserved. Third, GRPO achieves the highest score in the explicit thread across all three families, while DPO consistently falls below SFT_d, a pattern discussed in Section 5.2.

F Variety Classifier Accuracy Across Model Families

Table 8 reports variety classifier accuracy (%) for the explicit thread across all three model families, covering the four post-SFT stages. Accuracy is the proportion of the 25 evaluation prompt responses classified as the target variety. en-UK accuracy is near zero throughout all families and methods, confirming the classifier failure noted in Section 5.2 and Limitations; en-UK generation quality is therefore assessed via human and LLM-as-judge evaluation only. For en-AU and en-IN, DPO tends to maintain or improve classifier accuracy relative to SFT_d, while GRPO and GSPO are generally lower—consistent with the finding that higher feature density does not correspond to stronger variety classifier signal.

The en-UK variety classifier fails to generalise to LLM-generated outputs, with over 80% of responses classified as en-AU regardless of alignment method. This reflects a register distribution shift rather than a training failure: the classifier was trained on sentiment and sarcasm data rather than open-ended conversational responses. Accordingly, en-UK generation quality is not assessed via classifier accuracy and relies on human and LLM judgements instead.

Variety	Model	SFT_d	DPO	GRPO	GSPO
en-AU	Llama	**88.0**	**88.0**	84.0	76.0
	Qwen	**88.0**	76.0	80.0	76.0
	Gemma	**92.0**	84.0	80.0	80.0
en-IN	Llama	48.0	**64.0**	48.0	40.0
	Qwen	52.0	**76.0**	36.0	36.0
	Gemma	56.0	**60.0**	**60.0**	56.0
en-UK	Llama	0.0	**4.0**	0.0	0.0
	Qwen	0.0	0.0	0.0	**4.0**
	Gemma	**8.0**	0.0	0.0	0.0

Table 8: Variety classifier accuracy (%) for the explicit thread over 25 evaluation prompts. **Bold** indicates best result per model per variety. en-UK accuracy is near zero across all families and methods, reflecting classifier failure rather than genuine dialectal signal (see Section 5.2).

G Human Annotation

We recruited two native human annotators for each variety, with the expectation that each would spend less than 2 hours on the full set of tasks. They consented to the use of their annotation data for this research. Any annotators who were not authors of this work have been paid for their time. All annotators were male, except one female annotator for en-UK.

G.1 Task Instructions

We provided detailed instructions to our annotators on screen. The annotation process was divided into four tasks with 25 prompts per task.

> Overview: You will complete 100 judgements across 4 tasks (25 each). Each task asks you to read responses to a prompt and judge which sounds more dialectal for your assigned variety of English. Full instructions appear at the start of each task.

For Tasks 1 to 3, the instruction snapshot is quoted below.

> You will see 25 pairs of responses to the same prompt. Select whichever response sounds more dialectal for en-X English — closer to how a speaker of that variety might naturally respond. *Here X is replaced by the language variety.*

For Task 4, however, these were changed to:

> You will see 25 sets of three responses to the same prompt. Select whichever response sounds most dialectal for en-X English. There is no tie option—pick the best of the three. *Here X is replaced by the language variety.*

G.2 Generation Prompts

The following 25 prompts are used across all generation evaluation methods (automatic metrics, human evaluation, and LLM-as-judge). Prompts are manually curated to be open-ended, conversationally natural, and free of dialect-anchored geography, technical content, or sensitive topics.

Casual chat

- What's a good comfort meal after a rough day?
- What do you reckon makes a good night out with mates?
- What's your take on people who are always late?
- What are some good ways to wind down at the end of a long day?
- What's the best piece of advice someone's ever given you?

Opinion

- Do you think it's better to live in the city or the countryside?
- What do you think about people who just order takeaway every night instead of cooking?
- Is it worth spending a lot of money on a gym membership, or can you stay fit without one?
- What's the most overrated tourist destination in your opinion?
- Do you think social media does more harm than good?

Food & Lifestyle

- What's a simple meal worth recommending to someone just learning to cook?
- What's a good thing to eat on a lazy Sunday morning?
- Any tips for eating well without spending a fortune?
- What's a good breakfast to start the day right?
- What's a dish that everyone should try at least once?

Hypothetical

- If you could only eat one meal for the rest of your life, what would it be and why?
- If you had a free weekend with no plans or obligations, how would you spend it?
- If you could live anywhere in the world, where would you pick and why?
- If you could only keep three apps on your phone, which would you choose?
- If a friend asked you to help them move house, what would you bring to make the day easier?

Personal Reflection

- What's a hobby worth picking up, and what makes it enjoyable?
- What are some small things that can brighten up a rough day?
- What would an ideal Saturday look like from morning to night?
- What's a TV show or film worth recommending to everyone?
- What's a surprisingly useful thing to know that most people don't think about?

H Inter-Annotator Agreement

Inter-annotator agreement (Table 9) is modest overall, reflecting the inherent difficulty of perceptual dialectal judgements with a small annotator pool. Agreement is highest on average on Task 3 (AC_1 0.25–0.57; observed 48–64%), where raters consistently agree that the dialect-adapted output is more marked, consistent with the perceptibility finding in Section 5.3. Agreement is near chance on Tasks 1 and 4 ($AC_1 \leq 0.42$, mostly near zero), reflecting that the differences alignment introduces are small and hard to distinguish, consistent with the reward–quality gap. en-UK shows the strongest overall agreement (mean AC_1 0.44), including strong agreement on T2 (AC_1 0.66, 72% observed), the one task where raters showed stronger consensus; en-AU agreement is weakest (mean AC_1 0.02), consistent with the weak judgement results for that variety. Because several tasks are dominated by a single outcome, standard chance-corrected statistics are deflated by the prevalence paradox; we therefore report Gwet's AC_1 as the primary statistic alongside Krippendorff's α and Cohen's κ for transparency (Krippendorff, 2004; Gwet, 2008; Wongpakaran et al., 2013).

Variety	Task	AC_1	Kripp. α	Cohen's κ	Obs.%
en-AU	T1	−0.02	0.00	−0.02	32
	T2	−0.03	−0.18	−0.17	28
	T3	0.25	0.34	0.20	48
	T4	−0.11	−0.06	−0.06	24
en-IN	T1	−0.04	−0.14	−0.15	28
	T2	0.17	0.08	0.00	40
	T3	0.54	0.45	0.21	64
	T4	−0.25	0.01	0.00	12
en-UK	T1	0.42	0.36	0.36	60
	T2	0.66	0.43	0.27	72
	T3	0.57	−0.19	−0.08	64
	T4	0.11	0.24	0.16	40

Table 9: Inter-annotator agreement per variety and task. T1: Instruct vs SFT_d; T2: $GRPO_{all}$ vs $GRPO_d$; T3: SFT_d vs $GRPO_d$; T4: DPO/GSPO/GRPO forced choice. en-UK T3 illustrates the prevalence paradox: AC_1 0.57 despite α −0.19 and κ −0.08, owing to near-unanimous rater judgement in favour of SFT_d.

I Dialectal Generation: Linguistic Analysis

The reward signal and the variety classifier (Section 5.2) both derive from the same eWAVE feature space, so neither is an independent check on whether adapted models actually *produce* dialectal language. We therefore analyse the generation outputs with a separate, rule-based instrument that does not reuse the reward classifier. All measures are computed by a released script over the 1,375 evaluation outputs (three families × full pipeline × 25 prompts × variants).

Detectors. We detect three independent signal types: (i) *lexical* markers from curated per-variety lexicons (e.g. en-AU *arvo*, en-IN *prepone*, en-UK *nowt*); (ii) *orthographic* markers via explicit British/American spelling pairs (a classifier-independent en-UK/en-AU signal); and (iii) *morphosyntactic* features over spaCy POS/dependency parses—progressive-with-stative (*I am understanding*), *was/were* generalisation, possessive *me*, pluralised mass nouns (*informations*), bare adverbs, and invariant tags. Detectors are high-precision by design; reported rates are therefore lower bounds. From the firings we derive *density* (features per 1k tokens), *diversity* (distinct feature types), and *stacking* (features per sentence); the latter two are not captured by the density-based reward.

The reward–quality gap is corroborated by an independent measure, across families. Table 11 and Figure 2 show that, although GRPO maximises the eWAVE *reward* density (Table 7), it produces the *fewest* independent surface markers for Llama and Qwen (e.g. Qwen density 0.87 vs. SFT_d 2.34 per 1k tokens) and the lowest feature diversity; the ordering is mixed only for Gemma. These results suggest the eWAVE reward does not straightforwardly correspond to independently measured surface dialectal richness, extending the reward–quality gap (Section 5.2) beyond Llama to all three families on a non-circular measure.

Alignment redistributes, rather than introduces, dialectal marking. Figure 3 traces en-IN marker density along the pipeline: it is near zero through the base, CPT and instruct checkpoints, rises sharply at dialectal SFT, and is thereafter only shifted by the alignment methods, with GRPO (the method that most optimises the reward) sitting lowest. These results suggest surface dialectal marking is largely established by continual pretraining and supervised fine-tuning and subsequently redistributed by alignment, consistent with the robustness–generation gap (Section 5.2).

Human judgement tracks dialect, not reward density. Linking the measures to the human pairwise judgements on Llama (Table 12), annotators judge the higher-density output as more dialectal than the standard baseline (Task 1, Δdensity +1.38, 95% CI excludes 0) but judge the *more natural*, contracted output as more dialectal than the reward-maximising GRPO output (Task 3, Δcontractions +2.87; Δdensity negative, CI includes 0). The link between dialectal density and being judged more dialectal is therefore non-monotonic: features help against standard English, but their reward-driven maximisation does not, consistent with the gap reported in Section 5.2.

Surface marking is concentrated in en-IN. Table 10 shows that 37.7% of en-IN explicit-thread outputs carry a detected marker, against 10.0% for en-UK and 3.3% for en-AU; en-IN also shifts furthest from the standard pole. The varieties also differ in *which* features surface (Figure 4): en-IN marking is carried by mass-noun pluralisation and progressive-with-stative, whereas the few en-UK markers are predominantly possessive *me* and *was/were* generalisation. The en-AU markers we detect (bare adverbs, invariant tags) are essentially absent, consistent with the observation (Section 5.2) that Australian features are register-flexible and not reliably elicited by casual prompts.

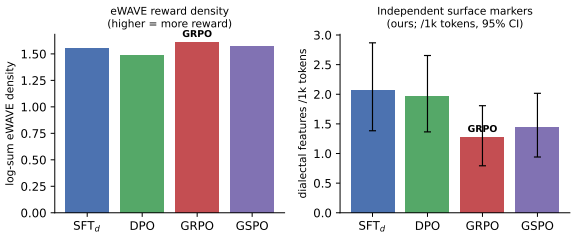


Figure 2: eWAVE *reward* density (left; reproduced from Table 7) versus *independent* surface-marker density (right; ours, bootstrap 95% CIs), averaged over the explicit thread. GRPO is highest on the reward it optimises yet lowest on independent markers—the reward–quality gap made visible.

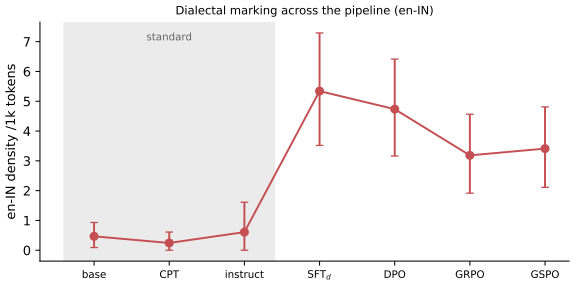


Figure 3: Dialectal-marker density across the pipeline for en-IN. Marking is near zero through base, CPT and instruct (standard region), rises sharply at dialectal SFT, and is thereafter only shifted by alignment—GRPO, the reward-maximising method, lowest (bootstrap 95% CIs).

Caveats. Detectors favour precision over recall, so densities are lower bounds; en-AU/en-UK are sparse and quantitative density claims are reliable mainly for en-IN; the human bridge is Llama-only with modest n (48–85 non-tie pairs per task), so deltas are reported with bootstrap 95% CIs as directional evidence. Code, detectors, and lexicons are released for reproduction.

Variety	% marked	Density (/1k)	JS vs. std.
en-AU	3.3	0.16	0.113
en-IN	37.7	4.17	0.361
en-UK	10.0	0.73	0.142

Table 10: Surface dialectal marking is concentrated in en-IN. % marked = share of explicit-thread outputs with ≥ 1 detected marker; JS = Jensen–Shannon divergence of the feature distribution from the standard pole (base/instruct), averaged over methods/families. en-AU markers (bare adverbs, invariant tags) are essentially absent, matching the register-flexibility note in Section 5.2.

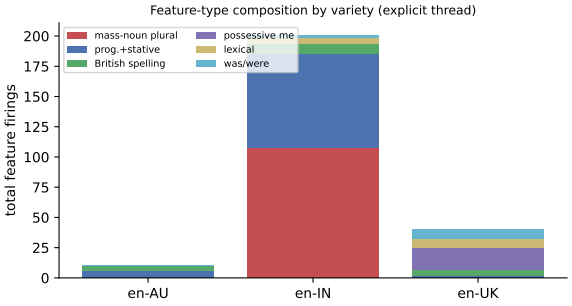


Figure 4: Feature-type composition by variety (total firings, explicit thread). en-IN marking is carried by mass-noun pluralisation and progressive-with-stative; en-UK by possessive *me* and *was/were* generalisation; en-AU markers are essentially absent.

Family	Method	Density (/1k)	Diversity (types)	Stacking (/sent)
Llama	SFT$_d$	1.51	0.19	0.028
	DPO	1.77	0.19	0.033
	GRPO	1.24	0.15	0.023
	GSPO	1.50	0.19	0.027
Qwen	SFT$_d$	2.34	0.25	0.043
	DPO	2.32	0.28	0.043
	GRPO	0.87	0.13	0.014
	GSPO	1.40	0.19	0.026
Gemma	SFT$_d$	2.36	0.21	0.042
	DPO	1.79	0.25	0.033
	GRPO	1.69	0.20	0.031
	GSPO	1.44	0.19	0.028

Table 11: Independent surface-marker measures on the explicit thread, averaged over en-AU/en-IN/en-UK (means over 25 prompts per cell). Density = curated dialectal features per 1k tokens; Diversity = distinct feature types; Stacking = features per sentence. In contrast to the eWAVE *reward* density (Table 7, where GRPO is highest), GRPO yields the *lowest* independent density and diversity for Llama and Qwen—the linguistic signature of the reward–quality gap (Section 5.2).

Task	ΔDensity	ΔStack	ΔContr.	n
T1: instruct vs SFT$_d$	**+1.38**	**+0.03**	-1.16	85
T2: broad vs targeted	+0.78	+0.01	-0.02	83
T3: SFT$_d$ vs GRPO$_d$	-0.51	-0.01	**+2.87**	48

Table 12: Human dialectal-judgement bridge (Llama). Mean winner−loser delta in each linguistic measure across non-tie pairwise trials; **bold** = bootstrap 95% CI excludes zero. Positive = the output judged more dialectal has *more* of the measure. n = non-tie trials for which both responses were available and the measure defined. T4 is excluded: a forced three-way choice has no winner−loser delta.